

Interpretations of numerals and structured contexts

Jae-Il Yeom
(Hongik University)

Abstract It has been assumed that the interpretation of a numeral n is determined as part of a scalar implicature, but there is a lot of evidence against it. First, numerals do not form a semantic scale but a pragmatic scale. Second, sometimes a numeral n is interpreted as ‘at least n ’ or ‘at most n ’, but the meaning of at least/most is not part of the meaning of a numeral. The meaning of a numeral is determined contextually. The basic principle is that we should make a statement as informative as possible. In making a statement informative, we consider two scales: one is the basic scalarity of largeness between numerals and the scalarity of likelihood. The two scales apply to two different regions with respect to a quantifier. The informativeness from the basic scalarity requires the numeral mentioned to be the maximal number of elements involved when each member in the quantifier domain is considered. And the alternative sentences with the numeral replaced with an alternative numeral are ordered by the informativeness from the unlikelihood scalarity, making the number mentioned a limit value among the maximal numbers of the elements involved in the meaning of the quantifier. The limit number n is taken to be ‘at least n ’ or ‘at most n ’. If no overt quantifier is involved, a universal quantifier over epistemic alternatives is provided by the support conditions of the sentence.

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1 Introduction

Grice (1961)’s maxim of quantity consists of the following two sub-maxims:¹

- Make your contribution as informative as is required for the current purposes of the exchange.

Address(es) of author(s) should be given

¹ The original maxim of quantity mentions the factor of relevance to the context. Later this factor is more or less ignored and informativeness is emphasized more, and the maxim has got other names like the Q-principle (Horn 1984 and Levinson 1987) or the principle of volubility.

- Do not make your contribution more informative than is required.

The maxims apply to utterances with scalar expressions, which include numerals. In this tradition, a numeral n is assumed to have the semantic meaning of ‘at least n ’. On the other hand, it gets the interpretation of ‘exactly n ’, because the speaker is supposed to follow the maxim of quantity: the speaker would mention a larger number if he or she could assert a stronger statement with a larger number, even if the number in the sentence really means ‘at least n ’. Therefore mentioning the number n means that the number is the largest one with which the speaker can make a statement truthfully. In the following example, the semantic meaning of (1a) is (1b) and (1c) is a scalar implicature. So we conclude that (1d) is what is intended by the speaker.

- (1) a. John has three children.
 b. John has at least three children.
 c. NOT(John has more than three children)
 d. John has *exactly three* children.

However, this way of reasoning is easily refuted by other statements with numerals. Cf. Horn (1992, 1996), Koenig (1991), Atlas (1992), etc. Numerals are different from other scalar terms:

- (2) A: Are many of your friends linguists?
 B: ?No, all of them are.
 B': Yes, (in fact) all of them are.
 (3) A: Do you have three children?
 B: No, four.
 B': ?Yes, (in fact) four.

We generally assume that if all of your friends are linguists, it is also true that many of your friends are. Thus when you make an affirmative answer to the question of whether many of your friends are linguists, you can claim that all of your friends are. But after you deny that many of your friends are linguists, you cannot very naturally claim that all of your friends are. In contrast, after you deny that you have three children, you can claim that you have four children, as shown in (3B). But after you assert that you have three children, it is awkward to claim that you have four children, as shown in (3B'). You may think of a situation in which the discourse becomes appropriate, but it is important that there are cases where the discourse can be inappropriate. This implies that a numeral n may not mean ‘at least n ’.

A more convincing example is like the following:

- (4) a. ??Neither of us liked the movie – she hated it and I absolutely loved it.
 (Horn 1996, 316)
 b. Neither of us have three kids – she has two and I have four.

Loving a movie entails liking it. If neither of us liked the movie, neither of us loved it. On the other hand, having four children does not entail having three, as (4b) shows.

This would be surprising if *three children* had the semantics of ‘at least three children’. This shows that numerals do not constitute a semantic scale.

A second argument against the semantic interpretation of a numeral *n* as ‘at least *n*’ is that a numeral *n* can have the ‘at most *n*’, ‘at least *n*’ or ‘exactly *n*’ interpretation, depending on the context. Sadock (1984, 143) and Carston (1998) gives some examples involving modality of possibility in which a numeral has an ‘at least’ or ‘at most’ interpretation:

- (5) a. That golfer is capable of a round of 100 (and maybe even 90/*110).
- b. Women with two pre-schoolers are eligible for the welfare benefit.

In (5a), the number 100 is interpreted as ‘at most 100’. This allows the possibility of a round of 90. In (5b), the number *two* has ‘at least’ interpretation because people with more pre-schoolers are more likely to get the welfare benefit.

When necessity is involved, a numeral tends to have ‘at least’ interpretation:

- (6) a. In Britain you have to be 18 to drive a car.
- b. Mary needs three A’s to get into Oxford.

The expressions *have to* and *need* involve the meaning of necessity, and the numerals indicate minimal requirements.

Breheny (2008) shows that the interpretation of a numeral is completely dependent on background knowledge:

- (7) No one who has three children is happy.

In (7), the quantifier *no one* introduces a downward-entailing context, but the numeral has three possible readings, depending on background knowledge. In a context where more children implies more stress, the numeral is likely to have ‘at least’ interpretation. In a context where more children implies more benefits, it is more likely to have ‘at most’ interpretation. Having four children is taken to be the minimal requirement to get some benefits. We can also think of a context where a specific number of children, say, three children for some reason, make their parents unhappy. In this context the numeral *three* is interpreted as ‘exactly three’. Breheny (2008) discusses this example to see if monotonicity affects the interpretation of a numeral, and it is shown that the main factor of determining the interpretation of a numeral is background knowledge.

One thing to note in (7) is that the example involves quantification. When there is no quantification or modality involved, a numeral generally has an ‘exactly’ or, less likely, ‘at least’ interpretation.

- (8) A: who has three children?
- B: John has three children.
- B’: John has three children, in fact he has five.

In this example, the numeral *three* seems to have an ‘exactly’ interpretation without the second sentence. But in B’'s response, it has ‘at least’ interpretation. This requires a context where having three children can be taken to be a minimal requirement for

something and we need someone who can satisfy the requirement. One interesting fact is that in neither case can we get ‘at most’ interpretation. We cannot think of a context where such an interpretation is accepted.

We have seen that whether modality and quantification are involved or not can affect the interpretation of a numeral. Despite this generalization, there is an example discussed by Krifka (1999) in which a numeral has an ‘at most’ interpretation even without any quantification.

- (9) In Guatemala, three percent of the population owns seventy percent of the land. (Krifka 1999)

In (9), there is no quantification involved. And the numeral *three* has an ‘at most’ interpretation, while *seventy* has an ‘at least’ interpretation. We have to be able to explain why numerals in general do not have ‘at most’ interpretations when there is no quantification involved. In addition, we have to explain why the numeral *three* in (9) has an ‘at most’ interpretation.

2 Previous analyses

The problems with the classical analysis of numerals have been discussed by many linguists, including Carson (1998), Breheny (2008), Van Kuppevelt (1996a, 1996b), Horn (1992, 1996), Koenig (1991), Atlas (1992), etc. Among these, Carson (1998) claims that the meaning of a numeral is *underspecified*:

- (10) X [THREE]
 , where the value of X is pragmatically determined. (cf. Recanati (1989))

In this interpretation, X can be ‘at least’, ‘at most’ or ‘exactly’, depending on the context the numeral is used. We have already seen that the interpretation of a numeral depends on the context where it is used, but simply saying that X is given by the context is not an explanation. We have to explain how the interpretation is determined and why modality or quantification affects the interpretation of a numeral.

Van Kuppevelt (1996a, 1996b) claims that the interpretation of a numeral also depends on whether it is in the topic or the comment part of the information structure of the utterance:

- (11) How many children does John have?
 a. He has three children.
 b. *He has three children, in fact five.
- (12) Who has three children?
 a. John has three children.
 b. John has three children, in fact he has five.

In (11), the numeral is in the focused part and gets the ‘exactly’ interpretation. This does not allow the possibility that John has more than three children. In (12), however, the numeral gets the ‘at least’ interpretation in the topic position of the sentence,

which allows the possibility that John has five children. These examples show that interpretation of numerals is affected by information structure.

However, this explanation has some problems. First, it is not explained when an ‘at most’ interpretation is possible. The ‘at most’ interpretation does not seem to depend on information structure. A second problem is that this explanation is not empirically supported either. Krifka (1999) refutes van Kuppevelt’s claim with the following example:

- (13) A: How many children does Nigel have?
 B: Nigel has fourteen_F children, perhaps even fifteen_F.

Even though the numeral *fourteen* is focused, it does not lead to ‘exactly’ interpretation: it still allows the possibility that Nigel has fifteen children.

Geurts (2006) gives a semantic analysis of numerals. He basically assumes that a numeral *n* is a predicate that applies to sum individuals, following Krifka (1999). In this usage, a NP with a numeral basically gets the ‘exactly’ interpretation:

- (14) These are five cows.
 (15) Five cows moored.

In the first example, *five cows* only means ‘exactly five cows’, while in the second example *five cows* means ‘exactly five cows’ or ‘at least five cows’.

He tries to explain an ‘at least’ interpretation by type-shifting, in terms of Partee (1986):

- (16) a. $\llbracket n \text{ CN} \rrbracket = \lambda P \exists! x [\#(x) = n \ \& \ \llbracket \text{CN} \rrbracket(x) \wedge P(x)]$
 b. $\lambda x [\#(x) = n \ \& \ \llbracket \text{CN} \rrbracket(x)]$ (Quantifier Lowering)
 c. $\lambda P \exists x [\#(x) = n \ \& \ \llbracket \text{CN} \rrbracket(x) \ \& \ P(x)]$ (Existential Closure)

He assumes that a numeral has the ‘exactly’ interpretation. Thus when a NP with a numeral is used as an argument, it is given the interpretation of (16a). By applying the rule of Quantifier Lowering to this, we can get the meaning of a NP with a numeral as a predicate, as given in (16b). And by applying Existential Closure to this, we can get the meaning of a NP with a numeral as an argument again, as given in (16c). This meaning is compatible with the situation in which *x* is more than *n*. This is how a numeral takes the ‘at least’ interpretation. By these derivations, Geurts tries to explain that a numeral can get an ‘exactly’ or ‘at least’ interpretation when a NP with a numeral is used as an argument, and that a numeral always has an ‘exactly’ interpretation when a NP with a numeral is used as a predicate.

Geurts explained the two meanings based on semantic type-shifting. However, one reading is missing: an ‘at most’ interpretation cannot be derived by semantic type-shifting. Instead, Geurts assumes that this interpretation is derived pragmatically, unlike the ‘at least’ interpretation. However, there is no evidence that ‘at least’ interpretations of numerals are semantically derived, while ‘at most’ interpretations of numerals are pragmatically derived. As will be shown below, interpretations of numerals are determined contextually. Moreover, even predicative uses of NPs with numerals can have ‘at least’ interpretations:

- (17) These are five pills, if not six.

We normally assume that *five pills* here means ‘exactly five pills’, but in this example it is more plausible to assume that it has the ‘at least’ interpretation. So a numeral in a predicate use does not necessarily have the ‘exactly’ meaning.

In this paper, I am going to account for how a numeral is interpreted, on the basis of a pragmatic approach. To support my analysis, I am going to make several claims that are opposite to what some previous analyses of numerals claim. First, I show that numerals do not constitute a semantic scale, and claim that the meaning of ‘at least/most’ is not part of the meaning of a numeral. This means that the interpretation of a numeral is determined contextually. There are two kinds of informativeness based on two kinds of scalarity: the basic scalarity between numerals and a contextual scalarity of likelihood. The informativeness from the basic scalarity requires the speaker or hearer to consider the maximal numbers of elements in a domain of quantification. The informativeness from the contextual scalarity requires the speaker to provide a limit value in the set of the maximal numbers of elements considered in the quantification domain, and the limit number n is taken to be ‘at least n ’ or ‘at most n ’. If no quantification is involved, quantification over epistemic alternatives is the default option.

3 Pragmatics in using numerals

3.1 Evidence against semantic scalarity

First, I will show that numerals do not constitute a semantic scale. This means that a numeral n does not entail ‘at least $n - 1$ ’ in general. One piece of evidence is that a set of alternatives among numerals is determined contextually. Suppose that John juggles with balls, and that it is the more difficult to juggle with more balls. And for some reasons, it is impossible to juggle with an odd number of balls.

- (18) A: Anyone who can juggle with more than six balls can participate in the competition. Does John juggle with six balls?

B: Yes, he does. He can even juggle with ten balls.

In this example, *six* means ‘at least six’, but B’s answer does not entail that John can juggle with five or seven balls. Note that the ‘at least’ interpretation does not have anything to do with semantic scalarity.

There is another example that supports that numerals do not constitute a semantic scale. Suppose that one boy ate three apples, and two boys ate two apples each (Krifka 1999).

- (19) a. Three boys ate seven apples.
b. Two boys ate six apples.

In this context, (19a) does not entail (19b). From the context, we can say either that two boys ate four apples or that two boys ate five apples, but not that two boys ate six

apples. Accumulation is not a semantic process. If numerals constitute a scale, the scalar set is determined in the context, not inherently in the language system.

Even though numerals constitute a pragmatic scale, there are many cases where they behave as if they constituted a semantic scale. This is related to the inclusion relation: for example, an event of reading six books includes an event of reading five books in general. But the relations between numerals can be restricted by the context: they are structured by the context. A sentence with a numeral '*five*' itself does not entail the corresponding sentence with the numeral '*six*'.

3.2 'At least/most' not part of semantics

In many of the previous analyses, the meanings like 'at least/most' are part of the meanings of numerals. However, there is evidence supporting that this is not actually the case. First, the same numeral may have different meanings depending on background knowledge. As I said, the interpretation of the numeral *three* in the following depends on which context the sentence is uttered in.

- (7) No one who has three children is happy.

The numeral *three* has three interpretations: 'exactly three', 'at least three', and 'at most three'. However, the numeral itself does not seem to have the meaning of 'at least' or 'at most'.

One piece of evidence is that the three interpretations are always available, whether or not the relevant scales are semantic or pragmatic. {love, like} is a set of semantic alternatives:

- (20) a. You must (at least) like studying maths if you want to do well in maths.
 b. You may (at most) watch her dance(, but you may not dance with her).
 c. No one who (at most) like studying maths do well in maths.
 d. No one who (at least) like studying maths fail in maths.

In these examples, the expressions in parentheses indicate the possible interpretations of *like studying maths*. It is implausible to assume that *like studying maths* itself has the meaning of 'at least/most like studying maths'. This shows that the meaning of 'at least' or 'at most' can be attached to any scalar term: it is not inherent part of the meaning of a numeral. It only comes from the context.

In (7), it is clear that directions of contextual scalarity are determined by *background knowledge* plus *the whole sentence*, and that contextual scalarity can be taken to be scalarity of informativeness: the less likely the utterance, the more informative it is. Suppose that more children implies more stress. Then ordering of being less likely is like the following with respect to numbers.

- (21) $1 > 2 > \boxed{3} > \dots > n-1 > n$

This means that the smaller the number, the stronger the sentence becomes pragmatically. When a number x is mentioned in the sentence, it means that the *lower* limit

number x such that no one who has x children is happy is three, so it gets the meaning of ‘at least’.

In the same background knowledge, however, the direction of scalarity changes if we change the sentence. Consider the following sentence.

(22) People who have three children are happy.

Since having more children implies more stress, the ordering of the sentence being less likely to be true is like the following:

(23) $n > n-1 > \dots > \boxed{3} > 2 > 1 > 0$

In this context the numeral *three* is taken to be the *upper* limit number such that people who have that number of children are happy. The numeral mentioned takes the meaning of ‘at most three’.

Now we consider the two sentences with different background knowledge. Suppose that more children implies more prosperity. Then the direction of scalarity changes in relation to sentence (7).

(24) Ordering of being less likely: $n > n-1 > \dots > \boxed{3} > 2 > 1 > 0$

Then the numeral gets the ‘at most’ interpretation. As for sentence (22), the direction of scalarity is reversed and the ordering of numerals is as follows:

(25) Ordering of being less likely: $1 > 2 > \boxed{3} > \dots > n-1 > n$

This indicates that the direction of scalarity changes with the whole utterance. Therefore the meanings like ‘at least’ and ‘at most’ must be associated with the whole sentence. So the meanings of the two sentences can be represented as follows:

- (26) a. AT_LEAST/MOST[No one who has three children is happy]
- b. AT_MOST/LEAST[People who have three children are happy]

That is, the numeral itself does not have the meaning of either ‘at most’ or ‘at least’, but rather the background knowledge plus the sentence as a whole determines the ordering of the (un)likelihood of the sentence. This, in turn, determines the meaning of the numeral by taking it as a limit value.

We can say a similar thing about sentences in which some kind of modality is involved. Suppose that people want to eat more apples. In this context, the numeral in the following sentence is interpreted as ‘at most three’.

(27) You may eat three apples.

This is a permission statement. If people want to eat more apples, it is less likely to give a permission to have more apples. So the ordering of being less likely is the following:

(28) Ordering of being less likely: $n > n-1 > \dots > \boxed{3} > 2 > 1$

In this respect, the larger the number, the stronger the permission becomes. So the number mentioned is taken to be the upper limit number such that the speaker can allow the hearer to eat that number of apples. This can be expressed as follows:

- (29) AT_MOST[You may have three apples]

Under the same background knowledge, the ordering of being less likely remains the same and sentence (30a) is understood as meaning (30b).

- (30) a. You must have three apples.
b. AT_MOST[You must have three apples]

Suppose that people want to solve fewer problems. Then it is less likely that permission or obligation statements are made with a smaller number. So the ordering of being less likely is like the following:

- (31) Ordering of being less likely: $1 > 2 > \boxed{3} > \dots > n-1 > n$

In this context, the following statements in (32) are interpreted as those in (33) respectively.

- (32) a. You may solve three problems.
b. You must solve three problems.
(33) a. AT_LEAST[You may solve three problems]
b. AT_LEAST[You must solve three problems]

4 Informativeness from the basic scalarity

I have discussed informativeness with respect to a pragmatic factor of being less likely. In some sense, uttering a sentence that is less likely to be true is uttering a pragmatically stronger sentence. This is the main factor that determines the meaning of a numeral. But in order for this account to make sense, we need to assume another scalarity. The traditional analysis of numerals is more concerned with a more basic kind of scalarity in numerals: that is, a larger number leads to a stronger statement than a smaller number. I have shown that there is no semantic scalarity among numerals, but pragmatically, an event of solving three problems includes an event of solving two problems in it, unless a case of solving two problems is excluded for some pragmatic reason.

- (34) a. John must solve three problems.
b. $\Box \exists x [\#(x) \geq 3 \wedge \text{problems}(x) \wedge \text{solve}(j,x)]$
c. John must solve two problems.
d. $\Box \exists x [\#(x) \geq 2 \wedge \text{problems}(x) \wedge \text{solve}(j,x)]$

The meaning of sentence (34a) can be represented as in (34b): pragmatically it is supposed to mean that John must solve at least three problems. That is, the number must be the lower limit of the number of problems John solves. On the other hand, if John solves three problems in a deontic alternative, it is also true that John solves two problems in the same alternative. We can say the same thing about other deontic alternatives. Then we should be able to utter (34b) truthfully. In the current context, however, we cannot say that John must solve two problems. This is problematic, considering the fact that in some deontic alternatives John solves four or five problems but (34a) is uttered.

There are two regions in each of which the maximality and the lower limit can be captured separately. Take sentence (34a) for example. It involves a deontic operator, a quantifier over a relevant set of deontic alternatives. When we talk about the lower limit of the numbers of problems John solves, we are considering only the maximal number of problems John solves in each deontic alternative. Here the basic scalarity comes in. If there is an event of John solving three problems in a deontic alternative, it is also true that there is a subevent of John solving two problems in that deontic alternative. But we only consider the maximal numbers of problems John solves in the deontic alternatives. If the number mentioned in (34a) is understood as the lower limit, it is among those maximal numbers based on the scalarity of (un)likelihood. In the interpretation of the numeral ‘3’, the basic scalarity of pragmatic inclusion applies to the number of problems John solves in each deontic alternative and determines the maximal number. The scale of unlikelihood orders alternative sentences with an alternative numeral, and the alternative numerals are taken to be the maximal numbers of problems John solves in each deontic alternative and determines the number mentioned as a limit value.

The same analysis can apply to cases of quantification over individuals. Consider (7) again. When more children means more stress, the sentence means that no one who have at least three children are happy. In the quantifier domain, we have people who have three children. A situation of having three children includes a sub-situation of having two children, but we only consider the maximal numbers of children the people in the quantification domain has. The number mentioned is taken to be the lower limit among the maximal numbers of children that people in the domain of quantification have.

The same distinction applies to cases where numerals have an ‘at most’ interpretation: the number mentioned is taken to be the upper limit, and it is the one in a set of maximal numbers.

(27) John may eat three apples.

$$\diamond \exists x[\#(x) \leq 3 \wedge \text{apples}(x) \wedge \text{eat}(j,x)]$$

(35) John may eat two apples.

$$\diamond \exists x[\#(x) \leq 2 \wedge \text{apples}(x) \wedge \text{eat}(j,x)]$$

(35) may follow from (27) pragmatically. Here again we only consider the maximal numbers of apples John eats in the deontic alternatives. 3 is the upper limit among them. The upper limit among the non-maximal numbers of apples people eat makes no sense. Since a situation of John eating four apples includes a situation of John

eating three apples, (27) could open up the possibility that John may eat four apples. We can say the same thing about (7) with the numeral 3 having the ‘at most n ’ interpretation.

The discussion so far indicates that the scalarity of unlikelihood applies to alternative sentences with an alternative numeral, whereas the basic scalarity among numerals themselves applies to each case in the quantification domain.

5 ‘At least/most’ interpretations and domains of quantification

Since the scalarity of unlikelihood applies to a set of alternative sentences that arise from alternative numerals in the sentence, the numeral itself does not have the meaning of ‘at least’ or ‘at most’. This claim can be supported by cases where the meaning of ‘at least/most’ cannot directly apply to numerals. In some cases, if a sentence with a numeral has the ‘at least/most’ interpretation, we get the same meaning we can get by putting the corresponding expression ‘at least/most’ before the numeral.

- (36) a. AT_LEAST[John must solve three problems]
 b. John must solve at least three problems.
- (37) a. AT_MOST[John may eat three apples]
 b. John may eat at most three apples.

Here (36a) and (37a) are considered to have the same meaning as (36a) and (37a), respectively. When the two meanings are the same, the scalar readings of the numerals are salient.

However, there are cases where this is not the case. Consider the following cases.

- (38) a. AT_MOST[John must eat *three* apples]
 b. ?John must eat at most three apples.
- (39) a. AT_LEAST[John may solve *three* problems]
 b. ?John may solve at least three problems.

In these two pairs, the second sentences do not seem appropriate. The semantic scope relationship like ‘AT_LEAST/MOST[... n ...]’ is always possible even when [...at least/most n ...] is not natural. This shows that even if a sentence with a numeral is interpreted with the meaning of ‘at least’ or ‘at most’, it does not mean that the numeral itself has the meaning of ‘at least’ or ‘at most’.

Now we have to explain why these differences are observed between (36-37) and (38-39). To do this, we need to explain first what it means that a sentence is interpreted with the meaning of ‘at least’ or ‘at most’. When a sentence includes a numeral, the use of a numeral introduces a set of alternative numerals, and hence a set of alternative sentences. If the original sentence includes a numeral in the scope of a quantifier, each alternative numeral is interpreted with respect to the domain of the quantifier in each alternative sentence. More specifically, each alternative numeral is interpreted with respect to a different domain of the quantifier, or with respect to a different nuclear

scope of the quantifier with the same quantification domain, depending on whether the numeral occurs in the restrictor or the nuclear scope.

The alternative sentences are ordered with respect to unlikelihood. If a numeral has the ‘at least/most’ interpretation in the restrictor or nuclear scope of a quantifier over the numeral, the restrictor or nuclear scopes of more likely alternative sentences can be added to that of the original sentence, with the numeral actually mentioned as a limit value. This idea allows us to explain the differences between (36-37) and (38-39). In (36a), *must* has narrower scope than AT_LEAST. This means that other alternative sentences with a larger number are more likely to hold. Then the nuclear scope of the quantifier *must* for the original sentence could be extended to cover the nuclear scopes of the alternatives with a larger number. The resulting meaning is the same as the meaning with the sentence with *three* replaced with *at least three*. This is why the numeral *three* is interpreted as meaning ‘at least three’.

In (38a), on the other hand, *must* has narrower scope than AT_MOST. This means that other alternative sentences with a smaller numeral are likely to hold. This means that the nuclear scope of the sentence actually uttered can be extended to cover the nuclear scopes of the alternative sentences with a smaller number. The resulting situation is that John eats three apples in some deontic alternatives and less than three apples in others. This is a situation in which the original sentence is false. In the resulting situation we could say that *John may eat three apples*, not that *John must eat three apples*. (38b) is odd just because there is no context in which the statement is appropriately uttered with *three* meaning ‘at most three’.

The intended meaning can be expressed with the following sentence.

- (40) John must eat three apples at most.

Here the expression *at most* is taken to be the operator equivalent to the operator of AT_MOST at the beginning of a sentence. So the sentence means that the upper limit of the numbers of apples John must eat is three.

A similar reasoning can apply to cases where a permission statement is combined with *at least*. In (37a), *may* has narrower scope than AT_MOST, which means that three is the upper limit of the numbers of apples John may eat. The nuclear scope of the sentence uttered can be extended to the nuclear scopes of the alternatives with a smaller number. This is the situation in which John eats three apples in some deontic alternatives and he eats less than three apples in other deontic alternatives. This is why we can say (37b) appropriately.

In (39a), on the other hand, *may* has narrower scope than AT_LEAST, which means that other alternative sentences with a larger number are more likely to hold. Thus the nuclear scope of the sentence uttered can be extended to cover the nuclear scopes of the alternative sentences with a larger number. Thus in some deontic alternatives John solves three problems and in some deontic alternatives John solves more than three problems. Still it is possible that in some deontic alternatives John solves less than three problems. This is problematic because *three* in this situation is not a limit value: the meaning of ‘AT_LEAST’ is lost. This is why (39b) is not an appropriate meaning.

In (41), a numeral occurs in the restrictor:

- (41) AT_LEAST/MOST[No one who has three_F children is happy]
 = No one who has at least/most three children is happy.

The restrictor can be extended to cases with a larger number or a smaller number. In this case the appropriateness of the resulting reading is not affected by the quantifier. It just has the effect of extending the domain of quantification. Thus AT_LEAST or AT_MOST at the beginning of the statement can be transferred to the interpretation of the numeral.

So far I have shown that in obligation statements numerals can have ‘at least’ interpretations, and that in permission statements, numerals can have ‘at most’ interpretations. If obligation statements have ‘at most’ meanings or permission statements have ‘at least’ meanings, we cannot put *at least/most* before numerals, because there is no appropriate situation in which such statements with *at least/most* before numerals can be uttered appropriately.

6 Statements with no quantification apparently involved

So far I have shown how numerals can get the meaning of ‘at least’ or ‘at most’ in the scope of a quantifier. However, a sentence with no quantifier involved can have the ‘at least’ interpretation in general, with some exceptional cases where a numeral get the ‘at most’ interpretation, as in (9). This implies that even in cases where no overt quantifier is involved, we need to assume that there is some kind of quantification involved. One such candidate is the universal quantification over epistemic alternatives. The quantifier comes from the support conditions of a statement:

- (42) ϕ is supported in an information state s iff $\forall w \in s \llbracket \phi \rrbracket^w = 1$.

A statement is supported in an information state iff it is true in every possible world in the information state. Thus all statements can be assumed to involve quantification over epistemic alternatives. We can also assume that the meaning ‘at least/most’ is related to lack of knowledge in some sense. Under the two assumptions, a statement uttered in a context is taken to be what the speaker can assert *at least* in the current information state. Thus the default ‘at least’ interpretation arises from epistemic necessity, just as obligation statements require numerals to have ‘at least’ interpretations.

Now consider (9) again. The statement is uttered in relation to the following pairs of alternative numerals.

- (43) In Guatemala, n % of the population owns m % of the land.

	←				
<i>strongest</i>	<1, 100>	<2, 100>	<3, 100>	<4, 100>	... <100, 100>
↑	<1, 99>	<2, 99>	<3, 99>	<4, 99>	... <100, 99>
	...	...	...	...	...
	<1, 71>	<2, 71>	<3, 71>	<4, 71>	...
	<1, 70>	<2, 70>	<3, 70>	<4, 70>	...
	<1, 69>	<2, 69>	<3, 69>	<4, 69>	...
	...	...	...	...	...
	<1, 1>	...	...	...	... <100, 1>
					<i>weakest</i>

The actual statement includes the pair $\langle 3, 70 \rangle$, where 3 is the upper limit and 70 is the lower limit, compared with other possible statements the speaker can assert. Then does the sentence mean that in Guatemala, at least n % of the population owns at most m % of the land? The answer is negative. This should be accounted for.

In my analysis, the statement has the following meaning:

- (44) $\Box$ [In Guatemala, 3 percent of the population owns 70 percent of the land]

Here $\Box$ is the universal quantification over epistemic alternatives. With the universal quantifier, the statement must be taken to be what can be asserted minimally at the current information state.

- (45) $\text{AT_LEAST}[\Box[\text{In Guatemala, 3 percent of the population owns 70 percent of the land}]]$

This opens up the possibility that the statement can be strengthened further if the speaker gets to know more. Since the statement focuses on the concentration of wealth to a small group of people. Thus in a stronger statement, n might be smaller and m might be larger. That is, there are some epistemic alternatives in the information state in which less than 3 percent of the population owns more than 70 percent of the land. So n has ‘at most’ interpretation, while m has ‘at least’ interpretation. The numeral n can have the ‘at most’ meaning because the two numbers are inversely related in the scale of alternatives.

We can apply the same reasoning to the following sentence, if it can be taken to be what can be asserted minimally at the current information state.

- (46) John ate three apples.
 $n > n-1 > \dots > \boxed{3} > 2 > 1$
 3 is the lower limit.

Even if the speaker can assert a weaker statement, the sentence does not mean that John ate at most three apples. The basic scalarity applies and we only consider the maximal number of apples John ate in each epistemic alternative. Applying the epistemic necessity to the statement, we can get the following meaning.

- (47) $\text{AT_LEAST}[\Box[\text{John ate three apples}]]$

With the universal quantifier over epistemic alternatives, *three* takes the ‘at least’ interpretation, unless the numeral has the ‘exactly’ meaning.

7 So-called ‘exactly’ interpretations

So-called ‘exactly’ readings obtain when all the other alternative numbers are excluded. There are two types of such cases. One is cases where scalarity lies in the background but it is known that other alternatives are not appropriate to assert, and the other is cases where the context is structured in a way that no scalarity is involved. In (48) the numeral *only three* is understood as having two meanings.

(48) John ate only three apples.

(48) means that John ate not more than three apples, or that the number of apples John ate is only three, not two or four. The former is a scalar reading, which is more plausible, and the latter is, less likely, a non-scalar reading. Even in the former case, we do not say that the numeral has ‘at least/most’ interpretation, because the speaker knows that John did not eat two or four apples. In neither case can we say that the numeral has different interpretations. This shows that so-called ‘exactly’ interpretations are obtained when the speaker knows that the other alternatives are excluded, regardless of whether numerals constitute scalar terms or not.

8 Conclusions

In this paper I have shown that numerals do not have ‘at least/most’ interpretations themselves. Such interpretations arise pragmatically when contexts provide such interpretations. Behind these interpretations lies directional likelihood with respect to numerals: the larger the numeral, the more or the less likely the whole sentence is to hold. In some sense, the less likely the statement, the more informative it is.

When we say that numerals constitute a scalar set, we are more concerned with the semantic strengths of them: something like ‘the event of eating three apples includes the event of eating two apples’, which has been the basic scalarity the previous literature on scalar implicatures has been more concerned with. But this plays a limited role in the interpretation of a numeral. The reason is that when we use a numeral, it is generally understood as the maximal numeral the speaker can say, unless the context is structured so that the maximal numeral is not relevant. When every numeral is relevant in the context, the interpretation of a numeral is determined by the informativeness from likelihood, by considering alternative numerals and alternative quantification domains for them. In cases where no explicit quantification is involved, we can assume that implicit universal quantification is involved over epistemic alternatives. This also can lead to some limit interpretation of the numeral mentioned in a statement, the lower limit in general.

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